



X(4th Sem.)-Computer Sc.-III(SEC-B-I)CBCS

2022

COMPUTER SCIENCE — HONOURS

Paper : SEC-B-I

(Information Security)

Full Marks : 80

The figures in the margin indicate full marks.

Candidates are required to give their answers in their own words as far as practicable.

Answer question nos. 1 & 2 and any four questions from the rest.

2/10

1. Answer *any ten* questions.

- ☒ (a) What do you mean by proxy server?
- ☐ (b) What do you mean by proxy firewall?
- ☒ (c) Which of the layers of OSI model ensure that data is successfully sent and received between two computers?
- ☒ (d) Briefly define field.
- ☐ (e) Write short note on steganography.
- ☒ (f) Define 'Worm'.
- ☒ (g) What is Block cipher?
- ☒ (h) Justify the use of prime numbers in RSA Algorithm.
- ☐ (i) Which topology is referred to as Hub-and-Spoke topology?
- ☒ (j) Give an example of DDoS attack.
- ☒ (k) What are the features of networking?
- ☐ (l) What is the functionality of Socket?
- ☒ (m) What is 'Trojan'?
- ☒ (n) What is Digital Watermarking?
- ☒ (o) What is called the Logic Bomb?

2. Answer *any four* questions.

5×4

- ☐ (a) Write short note on OS-hardening.
- ☒ (b) Distinguish between Block cipher and Stream cipher.
- ☐ (c) Distinguish between FTP and DNS server.

Please Turn Over

- ✓(d) Explain the concept of Information Security and describe its principles.
(e) State and prove Fermat's Little Theorem.
✓(f) Explain the Diffie-Hellman Key Exchange Algorithm.
✓(g) Explain the purpose of S-boxes in DES.
3. ✓(a) Describe S/MIME. 5+5
✓(b) How does the Firewall help to build a secure system?
- ✓4. ✓(a) What is a virus? What are its symptoms? (2+3)+5
✓(b) Explain P2P Network topology.
5. (a) Explain basic security threats.
(b) What are the Frameworks available for Network Management? – Explain. 5+5
- ✓6. ✓(a) Why SSL layer is positioned between application layer and transport layer? 5+5
✓(b) Name four key steps in the creation of a Digital Certificate.
- ✓7. ✓(a) What is Malware? What kind of precautions can be used to overcome the various problems?
✓(b) How can the TSR programs be used to override the ISR's of OS? Explain with an example. (2+3)+5
- ✓8. Write short notes on *any two* of the following : 5×2
(a) DHCP Server
(b) Hill cipher
✓(c) DNS Server
✓(d) Digital Signature
(e) Network Attacks.
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